

Muhammad Fazeel

Data Scientist. Large Language Model agents and applied deep learning

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PROFILE

Data scientist working on Large Language Models (LLMs) and applied deep learning. Two years supporting a flagship LLM product with a user base exceeding 500 million, focused on agentic behaviour, tool calling and evaluation, with systems shipped to production. Applied deep learning across medical imaging, digital pathology, genomics and physiological signals, and two live web platforms designed, built and deployed on Google Cloud Platform.

PROFESSIONAL EXPERIENCE

Data Scientist (Independent Contractor)

July 2024 to Present

Turing, Remote

- Contracted to a global technology client supporting a flagship Large Language Model (LLM) product with a user base exceeding 500 million.
- Optimise the model's agentic capabilities in Python, covering tool calling and multi-step autonomous task execution, working with the OpenAI API, LangChain, LlamaIndex and vLLM.
- Processed, evaluated and structured large-scale datasets, establishing the performance metrics used to judge the system's output quality.

Research Intern, Deep Learning (Master's thesis)

March 2026 to August 2026

GIPSA-Lab (CNRS) with CHU Grenoble Alpes and Inria, Grenoble, France

- Evaluated how far SegmentAnyMuscle, a muscle segmentation foundation model pretrained on adult MRI, transfers to pediatric MRI, and characterised its failure modes along acquisition, anatomical coverage and pathology on a clinical cohort of nine children.
- Built a human-in-the-loop annotation workflow in which clinicians corrected model predictions instead of redrawing them, producing 285 expert-reviewed labelled slices across 30 volumes from a previously unannotated archive.
- Benchmarked three annotation-efficient adaptation routes: parameter-efficient fine-tuning of the foundation model (frozen Vision Transformer backbone, trainable mask decoder, adapters and a mixture-of-experts head); a 2-D nnU-Net trained from scratch, which segmented a held-out severe-pathology thigh at Dice 0.94; and self-supervised masked-image pretraining on 123 unlabelled MRI series.
- Tested MedGemma, a medical vision-language model, as a controller for pixel-level annotation through tool-driven brush and eraser control and function-calling-driven segmentation, establishing where a general medical VLM stops being reliable for localisation.
- Designed the evaluation protocol for the degenerate case of an empty reference mask, replacing Dice with the muscle-to-total-tissue ratio and explicit false-positive accounting, delivered as a reproducible quantification module (PyTorch, nnU-Net v2, Hugging Face Transformers, LoRA, NVIDIA H100 on the GRICAD HPC cluster, GitLab).

AI Researcher

Nov 2022 to Aug 2024

Khyber Medical University, Peshawar, Pakistan

- Worked on a research grant co-funded by the Higher Education Commission (HEC) Pakistan and the British Council UK.
- Built deep learning pipelines on gigapixel whole-slide images to reduce inter-observer subjectivity in cancer grading.
- Engineered next-generation sequencing workflows with NVIDIA Clara Parabricks and delivered the downstream analysis that prioritised candidate cancer-associated genes for two published studies.

Research Intern

Nov 2021 to May 2022

AI in Healthcare Lab, National Center of Artificial Intelligence (NCAI), Peshawar, Pakistan

- Built deep learning classifiers for atrial fibrillation, bradycardia and tachycardia from ECG signals.
- Collaborated on sound source separation and audio denoising on spectrograms using generative adversarial networks, work that led to a published speech-enhancement study.

SELECTED PROJECTS

SpectraWeaver (agentic AI, open source, Python, github.com/fazeel15/SpectraWeaver)

Autonomous image-processing agent on Gemini 2.5 Pro: takes a stated objective, inspects each input image, writes the Python to achieve the goal, executes it in a constrained local runtime (NumPy, Pillow, matplotlib, scikit-image), and evaluates its own output against the objective before emitting a completion tag. Streamlit interface with branchable multi-turn conversation threads persisted as structured JSON.

Prodiasto (agentic AI, live web platform, Google Cloud Platform, prodiasto.com)

Designed, built and deployed an agentic AI services platform on Google Cloud Platform, including an LLM-powered conversational assistant for customer interaction.

Dilsho (full-stack, live web platform, Google Cloud Platform, dilsho.com)

Built and deployed a local-commerce progressive web app on Google Cloud Platform with live stock and price synchronisation across customer, merchant and rider interfaces.

Medical Model Studio (developer tooling, open source, Python and PyTorch, github.com/fazeel15/DLStudioGUI)

Desktop application for visually designing, validating and training PyTorch models on medical images: drag-and-drop network builder with live tensor-shape simulation, one-click export of a complete training pipeline, and an inspector that draws existing models as annotated diagrams.

Interpretable Broders' grading of oral cancer (digital pathology, deep learning)

Object-detection networks that detect and classify squamous cells in histopathology images and produce case-level Broders' grades backed by pathologist-interpretable evidence.

Sequencing pipeline for oral-cancer gene discovery (genomics, NGS data engineering)

Alignment, variant calling and annotation of raw Illumina FASTQ data with NVIDIA Clara Parabricks, then downstream analysis and visualisation to prioritise candidate oral-cancer-associated genes.

Arrhythmia classification on PhysioNet/CinC 2020 (physiological signals, deep learning)

Multi-class cardiac arrhythmia models with ECG preprocessing, class-imbalance handling and imbalance-aware evaluation metrics.

Sound source separation with coloured spectrograms (audio, generative models)

GAN-based denoising and source separation of audio mixtures, with waveforms encoded as coloured spectrograms to improve perceptual quality.

Defective column detection in infrared image sensors (image sensors and reliability, with LYNRED, Master's course project)

Statistical detection, classification and severity quantification of defective columns in IR sensor data for quality control, benchmarked on detection accuracy.

EDUCATION

Master's degree, Signal and Image Processing

2024 to 2026

Grenoble INP Phelma, Université Grenoble Alpes, Grenoble, France. Awarded September 2026

- Master 1: Data Analysis, Linking Experiments to Theory (DALETh). Master 2: Signal and Image Processing Methods and Applications (SIGMA). Both taught in English.
- Key modules: computational statistics and statistical learning, machine learning, image analysis and computer vision, image processing, bio and neuro imaging methods, signals and systems, scientific programming, data challenge.
- Thesis: Automatic Quantification of Skeletal Muscle in Pediatric MRI.

Graduate coursework, Robotics and Intelligent Machine Engineering

2023 to 2024

National University of Sciences and Technology (NUST), Islamabad, Pakistan

- Modules: artificial intelligence, computer vision, machine learning, mobile robotics, robot mechanics and control.

SKILLS

LLMs and agentic AI: Large Language Models (LLMs), Generative AI (GenAI), Natural Language Processing (NLP), agentic AI, tool calling, multi-step autonomous task execution, prompt engineering, LLM evaluation and benchmarking, Retrieval Augmented Generation (RAG), vector databases, model fine-tuning, LoRA, Model Context Protocol (MCP)

Deep learning and vision: PyTorch, TensorFlow, Hugging Face Transformers, nnU-Net, Vision Transformers, foundation model adaptation, parameter-efficient fine-tuning, self-supervised pretraining, semantic segmentation, object detection, generative adversarial networks, neural network design

Data and statistics: Python, NumPy, scikit-image, matplotlib, computational statistics and statistical learning, statistical analysis, dataset structuring and quality metrics, benchmark design, failure-mode and domain-shift analysis

Domains: MRI, whole-slide histopathology, ECG, audio and spectrograms, infrared image sensors, next-generation sequencing (Illumina, Oxford Nanopore), NVIDIA Clara Parabricks

Engineering and deployment: Google Cloud Platform (GCP), FastAPI, REST APIs, Docker, Git, GitLab, Streamlit, GPU training on HPC clusters (NVIDIA H100, GRICAD), VS Code, Cursor AI, LaTeX; also MATLAB and C++

Languages: English (C1, IELTS), Urdu, Hindko and Pashto (native), French (beginner)

CERTIFICATIONS

Certified Artificial Intelligence Developer (PIAIC); Applied Data Science with Python (Coursera Specialization); Neural Networks and Deep Learning (Coursera); Introduction to Programming with MATLAB (Coursera)

PUBLICATIONS

Exploring the mutational spectrum of key kinase genes PIK3CA, BRAF, EGFR, ALK and ROS1 in oral squamous cell carcinoma (2025). F. Nawab, W. Naeem, S. Fatima, A. Ali, A. T. Khalil, A. Mehmood, **M. Fazeel**, H. Ahmad, M. Alorini, M. Khan, I. A. Khan, M. Irfan, S. A. Khurram. doi:[10.1186/s12885-025-14609-8](https://doi.org/10.1186/s12885-025-14609-8)

Profiling genetic mutations in the DNA damage repair genes of oral squamous cell carcinoma patients from Pakistan (2025). W. Naeem, F. Nawab, M. T. Sarwar, A. T. Khalil, D. A. Gaber, H. Ahmad, **M. Fazeel**, M. Alorini, I. A. Khan, M. Irfan, M. Khan, S. A. Khurram, A. Ali. doi:[10.1038/s41598-025-91700-x](https://doi.org/10.1038/s41598-025-91700-x)

Single-channel speech enhancement using colored spectrograms (2024). S. Gul, M. S. Khan, **M. Fazeel**. doi:[10.1016/j.csl.2024.101626](https://doi.org/10.1016/j.csl.2024.101626)

VOLUNTEER WORK

Khyber Medical University, Advanced Center for Genomic Technologies: deployed accelerated computing pipelines that turn raw Oxford Nanopore and Illumina reads into consensus FASTA sequences. **UET Peshawar, Production and Automation Laboratory:** repaired and commissioned the EMCO Concept Mill 55 CNC machine, restoring it to service after 16 years.